

Exactness conditions for differentiable sliced optimal transport plans

Sliced optimal transport can recover not just a distance but an approximate transport *plan*, by lifting the closed-form one-dimensional solution back to the original space along an optimized projection. When this lifted plan coincides with the true optimal transport plan is not known in general. This internship seeks sufficient conditions — on the projector and the data — under which exactness holds, and validates them numerically.

§1. Organization

- **Start date:** As soon as possible (flexible).
- **Localization:** IRISA, Rennes.
- **Supervision:**
 - Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).
 - Laetitia Chapel (TAGADA research group, IRISA, Vannes).
 - Paul Viallard (MALT Inria, IRISA, Rennes).

§2. Context

Optimal transport (OT) yields both a distance between distributions and an optimal coupling — the transport *plan* — but computing the plan exactly costs $O(n^3)$. Sliced methods reduce the problem to one-dimensional projections, where OT is solved in closed form by sorting and the plan is a monotone permutation. Recent work lifts such a one-dimensional plan back to the original space and *optimizes* the projection — possibly a non-linear one parameterized by a neural network — to best approximate the full plan, casting the search as a differentiable bilevel problem (Chapel et al., 2025; Mahey et al., 2023). For generalized (non-linear) slicing to define a valid distance, the projection must satisfy injectivity-type conditions on the underlying generalized Radon transform (Kolouri et al., 2019; Bonneel et al., 2015).

These developments are largely *algorithmic*. The matching theoretical question — when the lifted, projector-optimized plan actually *equals* the true OT plan — has not been characterized, even though it determines whether the method is a principled approximation or merely a useful heuristic.

§3. Scientific challenge

The core object is a map from a projector to a lifted coupling, and exactness means this coupling equals an optimal π_{OT}^* . Two difficulties make this delicate. First, a sufficiently expressive projector class is needed for the optimum to be reachable at all, yet too expressive a class destroys the injectivity that guarantees a valid distance — so there is a tension between *expressivity* and *well-posedness* to be resolved. Second, π_{OT}^* need not be unique, so “recovering the plan” must be stated carefully (recovering *some* optimal plan, or the monotone one). The internship targets the constructive question: **what sufficient conditions on the projector architecture — and on the geometry of the two distributions — guarantee that the differentiable sliced plan coincides with an optimal transport plan?** A natural strategy is to start from cases where the answer is known (one dimension; Gaussians; distributions related by a monotone map) and to widen the class of distributions and projectors from there, verifying each step numerically.

§4. Goals

Concretely, the candidate will:

1. formalize the lifted-plan map and a precise notion of exactness that accounts for non-uniqueness of the optimal plan;
2. establish sufficient conditions for exactness in tractable settings (one-dimensional, Gaussian, or monotone-coupling cases) and identify which projector classes attain them while preserving validity of the distance;
3. build numerical diagnostics that measure the gap between the lifted plan and the true plan, and test the predicted conditions on controlled data;
4. (stretch / Ph.D. direction) begin a study of the optimization landscape of the projector search — characterizing critical points or benign geometry in a simplified setting — as a path toward guarantees that the differentiable solver actually finds an exact projector.

Candidates should possess:

- a strong mathematical background (analysis, linear algebra, convexity; notions of optimal transport are a plus),
- the ability to formulate and prove characterization results,
- good Python coding skills to validate the theory numerically,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis — the optimization-landscape question above is a natural thesis core. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a working command of the geometry of optimal transport and sliced approximations,
- experience proving exactness / recovery results and stating them rigorously,
- the habit of pairing theoretical conditions with numerical validation.

§5. Bibliography

- Bonneel, N., Rabin, J., Peyré, G., & Pfister, H. (2015). *Sliced and Radon Wasserstein Barycenters of Measures*. Journal of Mathematical Imaging and Vision.
- Chapel, L., Tavenard, R., & Vaiter, S. (2025). *Differentiable Generalized Sliced Wasserstein Plans*. NeurIPS. arXiv:2505.22049.
- Kolouri, S., Nadjahi, K., Şimşekli, U., Badeau, R., & Rohde, G. K. (2019). *Generalized Sliced Wasserstein Distances*. NeurIPS.
- Mahey, G., Chapel, L., Gasso, G., Bonet, C., & Courty, N. (2023). *Fast Optimal Transport through Sliced Wasserstein Generalized Geodesics*. NeurIPS.
- Peyré, G., & Cuturi, M. (2019). *Computational Optimal Transport*. Foundations and Trends in Machine Learning.